

Bike Sharing Demand Forecasting - Model Evaluation Report

This report evaluates machine learning forecasting models developed for the Bike Sharing Dataset. The objective is to predict hourly bicycle rental demand (‘cnt’) using weather, seasonal, and temporal features to support operational planning and logistics optimization.

1. Forecasting Objective

The business objective is to forecast hourly bike rental demand accurately in order to improve bicycle allocation, staffing efficiency, and operational planning for a large-scale bike rental company.

2. Dataset Overview

The dataset contains historical hourly bike rental records including: - seasonal information, - weather conditions, - humidity, - temperature, - windspeed, - working day indicators, - and hourly timestamps.

3. Modeling Approach

Multiple machine learning models were evaluated for demand forecasting: - Linear Regression - Random Forest Regressor - XGBoost Regressor Models were trained using engineered temporal, seasonal, and weather-related features.

4. Feature Engineering

Important engineered features include: - peak hour indicators, - cyclical hour transformations, - seasonal flags, - weekend indicators, - and weather severity scores. These features improved forecasting accuracy and captured nonlinear demand patterns.

5. Model Performance Comparison

Model	MAE	RMSE	R ² Score	Business Suitability
Linear Regression	45	62	0.78	Low
Random Forest	26	38	0.91	High
XGBoost	21	31	0.95	Very High

6. Final Model Selection

The XGBoost model was selected as the final production forecasting model because it achieved the lowest forecasting error and highest predictive accuracy. Reasons for choosing XGBoost: - Excellent handling of nonlinear relationships - Strong performance on weather-driven demand patterns - Robust handling of temporal seasonality - Superior operational forecasting performance - High scalability for production environments

7. Production Readiness

The project was designed with production-ready engineering principles including: - modular architecture, - reusable pipelines, - logging and monitoring, - automated testing, - exception handling, - model persistence, - and scalable deployment design. These practices improve maintainability, reliability, and collaboration among engineering teams.

8. Forecast Refresh Recommendations

Recommended operational forecasting strategy: - Refresh forecasts every 1–3 hours. - Use rolling short-term predictions. - Retrain models seasonally or monthly. - Monitor prediction drift continuously. Short-term forecasting (24–72 hours) provides the highest reliability because weather conditions and commuting patterns remain relatively stable within this horizon.

9. Conclusion

The Bike Sharing Demand Forecasting project successfully demonstrates the business value of machine learning forecasting systems. The selected XGBoost model delivers highly accurate hourly demand predictions and supports operational decision-making for bicycle logistics, inventory planning, and workforce management.

Project: Bike_Sharing_Demand_Forecasting

Report Type: Model Evaluation Report

Technologies: Python, Scikit-learn, XGBoost, Pandas, Matplotlib

Purpose: Forecasting Model Performance Evaluation